



DEALER BEST PRACTICE SERIES

Proper Gear Reminder for
Trucks Descending Grades
with ARC

Application Maintenance Site Management Component Rebuild Safety MARC Management

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with ARC1

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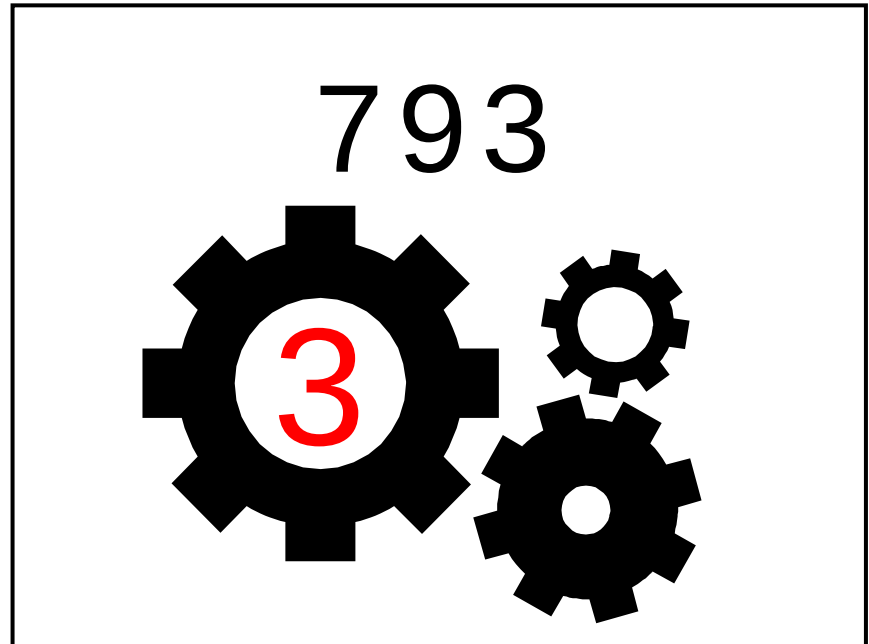
1.0 Best Practice Description

This Best Practice deals with reminding truck operators to select and engage the proper gear for descending a grade when using Automatic Retarder Control (ARC). For effective use of ARC, the truck must have the proper transmission gear engaged before descending the grade. To remind operators to make the gear change, mines have placed signs indicating the proper gear to be selected and engaged. The proper gear signs are placed near the beginning of the downhill grades. Each sign informs the operator which gear should be selected and engaged according to the machine's Operation and Maintenance Manual (O&MM).



A sign at Newmont Yannacocha in Peru. It reads:

For trucks with loads		
Temperature of the brake oil:		
110 - 121 Deg C		
Truck	Gear	Speed (kph)
793	4	27
785	4	31
777	5	34



A suggested proper gear sign for a 793

2.0 Implementation Steps

- Establish the proper gear to be selected and engaged for each grade in the mine as per the O&MM.
- Train operators
- Place the correct gear indicator signs prior to each downhill grade.
- Additional signs may be advisable along the grade to remind the operator of the correct gear.

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3.0 Benefits

Proper gear selection while descending grades allows for effective use of the Automatic Retarder Control. ARC is designed to reduce brake wear and brake system overheating. Many operators overlook the fact that the transmission must be engaged in a particular gear suitable for descending the grade. Without the proper lower gear, the ARC will not be effective. Engine overspeed, accelerated brake wear, and brake system overheating can result.

4.0 Resources Required

All that is required are signs appropriately placed before downhill grades that inform the operator of the proper gear to use while descending the grade. The sign may be as simple as a symbol of a gear with a number at the center. The number indicates the correct gear to be selected and engaged before descending the grade. Such a pictograph avoids any language problems. In situations where different truck models require different gears to be selected and engaged, the sign may indicate the particular truck model: 793, 797, etc.

5.0 Supporting Attachments

None applicable.

6.0 Related Best Practices

None applicable

7.0 Acknowledgements

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